Reg. No. | | | | | | | | | | | [13ME654/12ME654]

SIXTH SEMESTER B.Tech./B.E. EXAMINATION, JUNE – 2016

SOLID MECHANICS

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data, if any, may be suitably assumed.

1. **a.** Explain the assumptions made in linear elasticity. **04**

b. Mention the applications of linear elasticity. **02**

c. Derive the equations of equilibrium for 2-D stress state. **08**

2. **a.** The state of stress at a point is given below **06**

$$\begin{bmatrix} 100 & 80 & 0 \\ 90 & -60 & 0 \\ 0 & 0 & 40 \end{bmatrix} KPa$$

Considering another set of co-ordinate axis $x'y'z'$ in which z' coincides with z and x' is rotated by 30^0 anticlockwise from x-axis. Determine the stress components in the new co-ordinate system.

b. The stress component at a point in a body are given by **08**

$\sigma_x = 3xy^2z + 2x$ $\tau_{xy} = 0$

$\sigma_y = 5xyz + 3y$ $\tau_{yz} = \tau_{xz} = 32xy^2z + 2xy$

$\sigma_2 = x^2y + y^2z$

Determine whether these components of stress satisfy the equilibrium equations or not at a point (1, -1, 2). Determine suitable body force vector required at this point so that these stress components are in equilibrium with the external force.

3. **a.** Derive the equations of compatibility for strain in three dimensions. **09**

b. What are rectangular strain components? Write the expressions for the same. **05**

4. **a.** What are elastic constants? Explain generalized Hooki's law. **06**

b. Explain uniqueness theorem and saint Venant's principle. **08**

5. Derive the expressions for stresses for bending of a narrow cantilever beam of length L and rectangular cross section subjected to end load. **14**

6. a. What is axisymmetric? Explain with an example. **05**

b. The internal and external diameter of a thick hollow cylinder is 80mm and 120mm. it is subjected to an external pressure of $40MN/m^2$ when the internal pressure is $120MN/m^2$. Calculate circumferential stresses at external and internal surface and also at mean radius. Draw the stress distribution. **09**

7. a. Explain membrane analogy with an example. **04**

b. A thin walled box section having dimensions 2a x a x t is to be compared with a solid circular section of diameter as shown in Fig. 7(b). Determine the thickness 't'. so that the two sections have. **10**

i. Same maximum shear stress for the same torque.

ii. Same stiffness

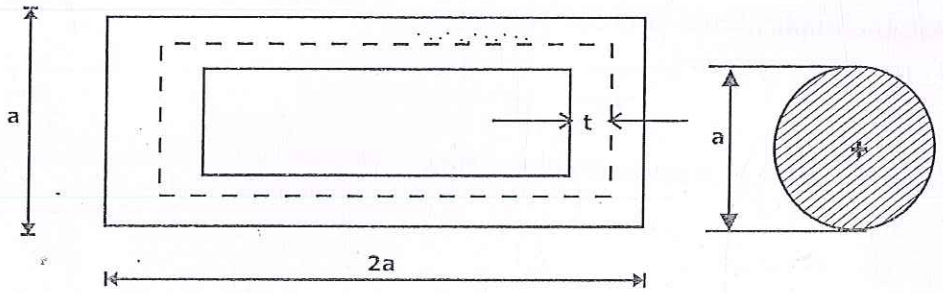

Fig. 7(b)

8. a. What is yield criteria? Explain Tresca yield criteria. **05**

b. Explain Taylor and Quinney's experiments in support of yield criteria. Mention the conclusions drawn from the above experiments. **09**